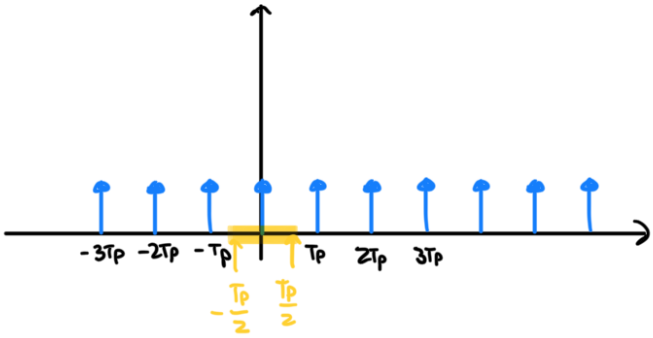
Lezione 13 - 4/04/2024

ESERCIZIO 1a

$s(t) = \text{comb}_{T_p}(t) = \text{rep}_{T_p}\,\delta(t)$ (treno di impulsi)



- $S_K = ?$
- SERIE DI FOURIER?

SOL. USO LA DEFINIZIONE DI S_K

$$\boxed{S_K = \frac{1}{T_p}\int_{t_0}^{t_0+T_p} s(t)\, e^{-jK\omega_0 t}\, dt} \qquad \omega_0 = \frac{2\pi}{T_p}$$

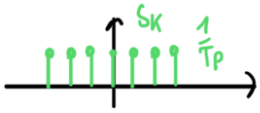
CI METTIAMO IN UN INTERVALLO CENTRATO IN 0 DI AMPIEZZA T_p

$$S_K = \frac{1}{T_p}\int_{-\frac{T_p}{2}}^{\frac{T_p}{2}} \delta(t)\, e^{-jK\omega_0 t}\, dt = \frac{1}{T_p}\, e^{-jK\omega_0 \cdot 0} = \frac{1}{T_p}\cdot 1 = \frac{1}{T_p}$$

$$\Rightarrow \boxed{s(t) = \text{rep}_{T_p}\,\delta(t) \xrightarrow{\mathcal{F}} S_K = \frac{1}{T_p}}$$

delta periodico

segnale costante

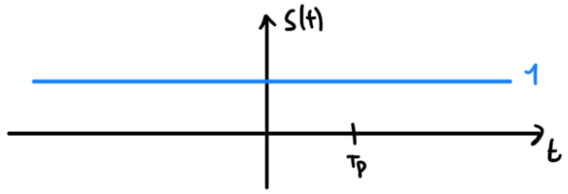


$$s(t) = \sum_{K=-\infty}^{+\infty} S_K\, e^{jK\omega_0 t} = \underbrace{\frac{1}{T_p}\sum_{K=-\infty}^{+\infty} e^{j2\pi K \frac{t}{T_p}}}_{\text{SERIE DI FOURIER}}$$

converge alla $\text{rep}_{T_p}\,\delta(t)$

ESERCIZIO 1b (scrivete l'esercizio 1a inverso)

$s(t) = 1$

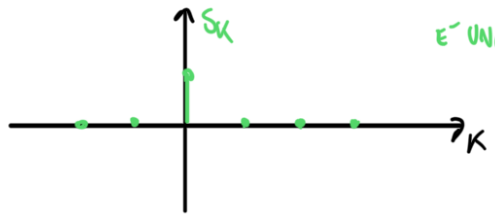


$$S_K = \frac{1}{T_P}\int_0^{T_P} 1\, e^{-jK\omega_0 t}\, dt \qquad \omega_0 = \frac{2\pi}{T_P}$$

ricorda: questo è un esponenziale complesso a fase lineare, cioè una sinusoide

$$= \begin{cases} \frac{1}{T_P}\cdot T_P = 1 & K = 0 \\ 0 & K \neq 0 \end{cases}$$

$$\frac{1}{T_P}\left[\frac{e^{-jK\omega_0 t}}{-jK\omega_0}\right]_0^{T_P} = \frac{e^{-jK\omega_0 T_P} - 1}{-jK\omega_0 T_P} = \frac{1\cdot 1}{-jK\omega_0 T_P} = 0$$



È UNA $\delta(K)$!

$$s(t) = 1 \xrightarrow{\mathcal{F}} S_K = \delta(K)$$

SEGNALE COSTANTE

DELTA DISCRETA

ABBIAMO SCOPERTO CHE C'È UNA DUALITÀ TRA DELTA E SEGNALI COSTANTI

$$\boxed{\text{delta} \xleftrightarrow{\mathcal{F}} \text{SEGNALE COSTANTE}}$$

DUALITÀ DELTA - COSTANTE

$$s(t) = \sum_{K=-\infty}^{+\infty} S_K e^{jK\omega_0 t} = 1$$

VALORE MEDIO E POTENZA (alla luce dei risultati delle slide 25 e 26)

a) $s(t) = rep_{T_p}\,\delta(t) \xrightarrow{\mathcal{F}} S_K = \frac{1}{T_p}$

$$m_s = S_0 = \frac{1}{T_p}$$

$$P_s = \sum_K |S_K|^2 = \sum_{K=-\infty}^{+\infty} \frac{1}{T_p^2} = \infty$$

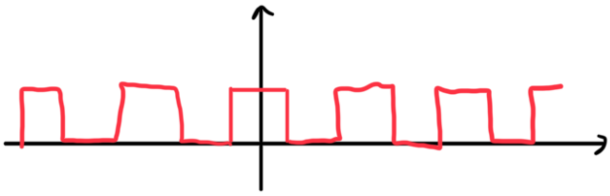
b) $s(t) = 1 \longrightarrow S_K = \delta(K)$

$$m_s = S_0 = 1$$

$$P_s = \sum_K |S_K|^2 = 1$$

c) $s(t) = rep_{T_p}\, rect\left(\frac{t}{2a}\right) \longrightarrow S_K = d\,\mathrm{sinc}(Kd) \qquad d = \frac{2a}{T_p}$

$$m_s = S_0 = d$$



$$P_s = \sum_{K=-\infty}^{+\infty} |S_K|^2 = \sum_{K=-\infty}^{+\infty} d^2\,\mathrm{sinc}^2(Kd) = ?$$

(questa serie non è risolvibile)

$= d$ (ma calcolandolo dal dominio del tempo)

- il segnale **costante** s(t) = 1
- la **sinusoide** $s(t) = A\cos(\omega_0 t + \varphi_0)$ con $\omega_0 = 2\pi/T_p$
- il **segnale** $s(t) = x(t)\cos(10\omega_0 t)$ con $\omega_0 = 2\pi F_0$, $F_0 = 1/T_p$ e T_p la

ESERCIZIO 1c

$s(t) = \cos(\omega_0 t + \varphi_0)$ $\qquad \omega_0 = \frac{2\pi}{T_p}$

a) $S_K = ?$
b) $m_s = ?$
c) $P_s = ?$

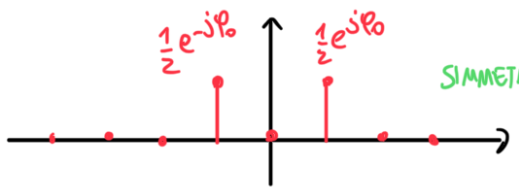
Sol. a. SCRIVO IL COSENO SOTTO FORMA DI ESPONENZIALI COMPLESSI CON EULERO

$$s(t) = \underbrace{\frac{e^{j\varphi_0}}{2}}_{S_1} e^{j\omega_0 t} + \underbrace{\frac{e^{-j\varphi_0}}{2}}_{S_{-1}} e^{-j\omega_0 t}$$

$K=1$ $\qquad K=-1$

$$= \sum_K S_K e^{jK\omega_0 t}$$

$$S_K = \begin{cases} \frac{e^{j\varphi_0}}{2} & K=1 \\ \frac{e^{-j\varphi_0}}{2} & K=-1 \\ 0 & \text{ALTROVE} \end{cases}$$



SIMMETRIA HERMITIANA: $S_{-K} = S_K$

ABBIAMO AGITO PER ISPEZIONE: abbiamo guardato la struttura del segnale e abbiamo visto che era già espresso sotto forma di SDF con 2 coefficienti attivi

b) $m_s = S_0 = 0$

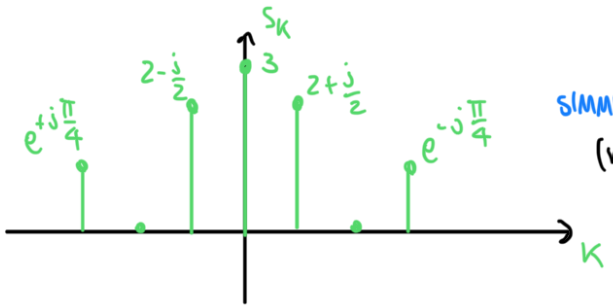
c) $$P_s = \sum_K |S_K|^2 = \left|\frac{e^{j\varphi_0}}{2}\right|^2 + \left|\frac{e^{-j\varphi_0}}{2}\right|^2 = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

ESERCIZIO

$$s(t) = \underset{0\cdot\omega_0=0}{3} - \sin(\underset{\omega_0=2}{2}t) + 4\cos(2t) + 2\cos(\underset{3\omega_0}{6}t - \frac{\pi}{4})$$

$$s(t) = 3 - \frac{e^{j\omega_0 t} - e^{-j\omega_0 t}}{2j} + 4\,\frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2} + 2\,\frac{e^{j3\omega_0 t}e^{-j\frac{\pi}{4}} + e^{-j3\omega_0 t}e^{j\frac{\pi}{4}}}{2}$$

$$= 3 + \left(\frac{j}{2}+2\right)e^{j\omega_0 t} + \left(-\frac{j}{2}+2\right)e^{-j\omega_0 t} + e^{-j\frac{\pi}{4}}e^{j3\omega_0 t} + e^{j\frac{\pi}{4}}e^{-j3\omega_0 t}$$



SIMMETRIA HERMITIANA: $S_{-K} = S_K^*$

(regola di controllo)

$M_s = 3$

$$P_s = 3^2 + 2\left|2+\frac{j}{2}\right|^2 + 2\left|e^{-j\frac{\pi}{4}}\right|$$

$$= 9 + 2\left(4+\frac{1}{4}\right) + 2\cdot 1 = 9 + 8 + \frac{1}{2} + 2 = 19 + \frac{1}{2} = 19{,}5$$